SpeechSaw

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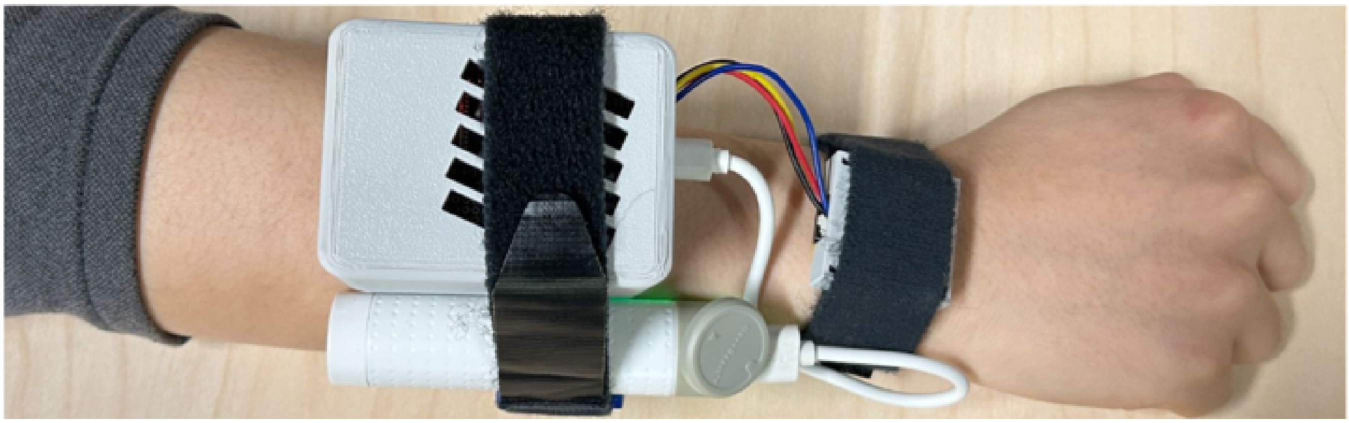

Figure 1: SpeechSaw prototype: wrist-mounted VPU module with ESP32S3 and Battery

Abstract

Conventional air-conductive microphones collect ambient sound and the speaker's voice indiscriminately, yielding high transcription error rates in crowded streets, hiring fairs, or multi-party meetings, and raising privacy concerns when by-standers' speech is inadvertently recorded.[1] To address these limitations we introduce *SpeechSaw*, a wearable speech-capture system that employs a Vibrational Processing Unit (VPU) to sense bone- and skin-borne vibrations produced exclusively by the wearer's vocal apparatus.[2] Because vibration signals are largely immune to airborne noise, *SpeechSaw* can isolate the target speaker without directional beamforming or multi-mic arrays, while inherently suppressing bystander speech and thus mitigating privacy leakage.

Our prototype integrates a Sonion voice pick-up VPU with an ESP32–S2 micro-controller that converts the raw PDM vibration stream to PCM and relays it via Wi-Fi to a host PC running the *Whisper* large-vocabulary speech recogniser.[3] The end-to-end pipeline delivers real-time transcription with speaker identification.

Empirical evaluation shows that, in quiet conditions, the Bone-VPU matches the word-error rate (WER) of a studio-grade microphone. In the presence of competing talkers or environmental noise, it achieves $\approx$ 30% lower WER than the microphone baseline, demonstrating superior robustness to interference.

These results indicate that vibration-domain sensing is a compelling alternative for high-fidelity, privacy-preserving speech capture in both noise-challenged and privacy-sensitive scenarios, paving the way for next-generation wearable voice interfaces.

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CCS Concepts

• **Hardware** → **Sensor devices and platforms**; *Sensor applications and deployments*; • **Human-centered computing** → *Sound-based input / output*.

Keywords

VPU, wearable sensing, speech recognition, privacy, bone conduction

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1 Introduction

Voice interaction is increasingly permeating everyday life—from hands-free voice messages sent on crowded city streets to remote recruitment interviews conducted in reverberant halls. Yet the effectiveness of these interactions hinges on the fidelity with which a system captures the user's speech. Conventional air-conductive microphones cannot distinguish among different speakers or separate environmental noise; consequently, ASR pipelines that rely on them suffer sharply elevated word-error rates (WER) when multiple people speak, public-address systems blare, or traffic roars.[10] Worse still, such microphones may inadvertently record by-standers' speech, raising privacy concerns and exposing service providers to ever-stricter legal regulations.

Why existing solutions fall short. Prior work has pursued two main ways. *(i) Beam-forming arrays* spatially filter unwanted sources, but demand precise array calibration and deteriorate when interfering speakers are co-located with the user.[9] *(ii) Learned noise-suppression networks* can suppress diffuse noise, yet incur additional latency and cloud-compute cost.[12] Consequently, real-time, noise-robust and privacy-preserving speech capture for mobile users remains an open challenge.

Our insight. Airborne acoustics are highly contaminable, whereas the vibrations propagating through a speaker's bones and soft tissue are intrinsically private and resilient to external sound. Recent advances in miniaturised piezoelectric transducers—commercialised as *Vibrational Processing Units* (VPUs)[5]—enable millivolt-level detection of such tissue-borne signals in a watch-class footprint. Leveraging this opportunity, we present *SpeechSaw*, the wearable system that replaces—or augments—the conventional microphone with a wrist-mounted VPU and end-to-end software stack for low-latency transcription.

System overview. SpeechSaw couples a Sonion voice pick-up VPU to an ESP32-S2 micro-controller[6] that time-multiplexes pulse-density-modulation streams from both the VPU and a reference microphone on a single I^2S interface. The MCU pre-filters, converts PDM to PCM, and relays packets via Wi-Fi to a host PC running the *Whisper* large-vocabulary ASR engine;[4] a multi-threaded pipeline yields live transcripts and speaker identification with < 2 s end-to-end latency.[11]

2 System Overview

SpeechSaw converts tissue-borne vibrations into textual transcripts in real time. Fig. 2 sketches the end-to-end pipeline, which starts with a wrist-mounted VPU and ends with on-screen text and speaker identity.

- **Input.** A dual-channel pulse-density–modulated (PDM) audio stream comprising: (i) bone-conducted vibrations from the VPU and (ii) an optional reference microphone.
- **Output.** The recognised text and the detected speaker ID are pushed to a Web front-end with < 2 s median latency, enabling live captioning, note-taking and archival.[13]

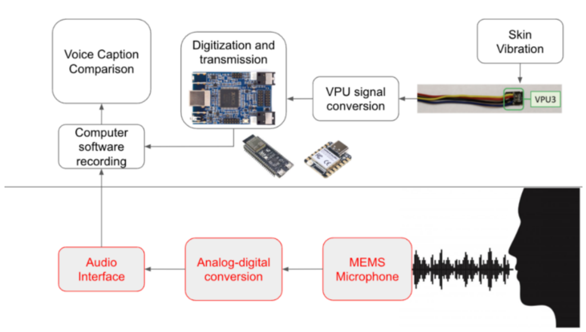

Figure 2: End-to-end SpeechSaw pipeline.

This high-level view provides the context for the hardware iterations detailed next in Section 3, and it sets the stage for the signal-processing and real-time ASR software described later in Section 4.

3 Hardware Design & Implementation

This section walks through three hardware iterations **MCHStreamer, ESP32S3, and ESP32S3 XIAO** we tried and trade-offs at each stage. A summary table comparing the three prototypes closes the section.

3.1 MCH Streamer Prototype

3.1.1 Overview. Our initial prototype relied on a USB *MCH Streamer*. Both the VPU and the reference microphone were wired to the board, which was then attached to a host PC; *Audacity* allowed us to visualise the captured PCM waveforms in real time. Because the MCH Streamer performs the PDM → PCM conversion in hardware, the electrical setup remained straightforward.

3.1.2 hardware setup. Figure 3 illustrates the wiring of the VPU and the reference microphone to the USB *MCH Streamer*.

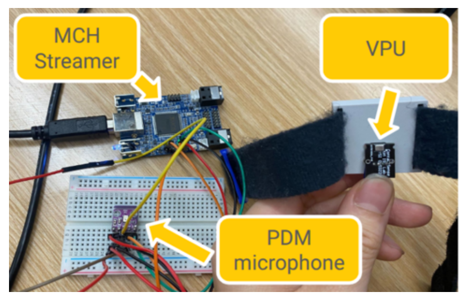

Figure 3: hardware setup of the MCH Streamer–based prototype.

3.1.3 Take Away. The key takeaway from this version was that we could successfully record audio from the VPU and the reference mic simultaneously, while gaining hands-on familiarity with the PDM, PCM, and I^2S protocols.

3.2 ESP32-S3 Prototype

3.2.1 Overview. We first evaluated the `ESP32-C6` because of its compact footprint, but the data sheet revealed that the chip supports only *PCM-to-PDM TX* and lacks the inverse *PDM-to-PCM RX* path required to ingest raw sensor data. We therefore migrated to the slightly larger yet feature-complete `ESP32-S3`, whose integrated I^2S peripheral offers full-duplex PDM support.

The VPU and the reference microphone share a single data line and clock line:

- the VPU samples on the *falling* clock edge.
- reference microphone samples on the *rising* edge.

enabling seamless time-multiplexing.

This revision also implements Wi-Fi transport. Both the transmitter (ESP32-S3) and the receiver (host PC) run a lightweight *loss-bag detector*: each 1 024-byte audio packet carries a sequence index; if the receiver detects a gap, it issues a `RESEND` request for the missing index, thereby guaranteeing a continuous audio stream.

The ESP32-S3 allocates a 1 024-byte ring buffer for each channel; once the buffer is filled, the firmware dispatches the entire packet over 802.11 n in a single transaction.

Data-bag structure. Each 1 024-byte audio bag contains a 2-byte header, 1024-byte audio data, and a 1-byte sequence counter, as illustrated in Figure 5.

3.2.2 hardware setup. See below.

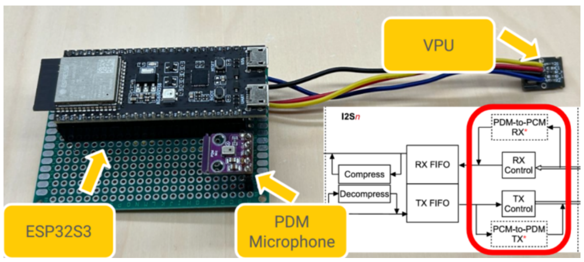

Figure 4: hardware setup of the ESP32S prototype.

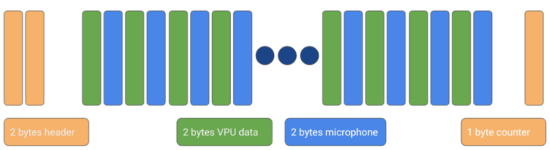

Figure 5: Layout of the 1 024-byte audio bag used by the Wi-Fi link.

3.2.3 Take Away. We verified the reliability of Wi-Fi transmission on the ESP32-S3. In addition, we tested audio signals at different frequencies (see Fig. 11 Fig. 12 Fig. 13) to demonstrate the stability of the signal flow. Meanwhile, we integrated this hardware with the original Whisper pipeline and successfully achieved real-time speech-to-text conversion.

3.3 XIAO ESP32-S3 Miniature Prototype

3.3.1 Overview. For our final prototype, we transitioned to the XIAO ESP32-S3 form factor for improved portability and integration. The design employs two PDM signal sources—a traditional PDM microphone and a VPU with PDM output—sharing the single PDM_DATA and PDM_CLK pair available on the XIAO board through select pin toggling.

All PCB and peripheral circuitry were designed based on the schematic and layout of the XIAO ESP32-S3. The S3's onboard battery management chip allowed us to omit external power-management circuitry. We created several PCB layouts around the S3's footprint, with the current design prioritizing signal integrity and portability. By adopting the DF40C-30DP connector, the second-generation PCB achieved a 25% reduction in area compared to the first version.

3.3.2 hardware setup. Figure 6 illustrates the XIAO ESP32-S3 hardware design, highlighting the front-mounted microphone and rear-positioned VPU in the PCB layout. Furthermore, in our final design, we created a 3D model to contain the prototype board with ESP32S3 (Figure 9) and the VPU (Figure 10).

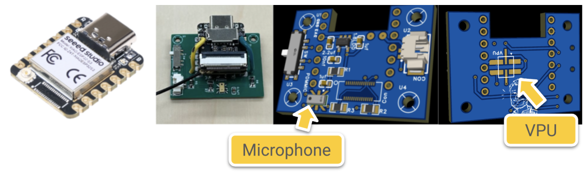

Figure 6: Figure of the XIAO ESP32-S3 board and PCB design.

3.3.3 Take Away. A significant challenge emerged with the 1.8 V LDO overheating, potentially due to excessive load current or soldering issues. To address this, we replaced our PDM microphone with a 3.3 V variant powered directly from the ESP32's internal 3.3 V LDO, reducing overall power consumption.

4 Software Pipeline

SpeechSaw runs three co-operating software threads that together form a real-time, end-to-end pipeline (Figures 7 and 8). All threads are lock-free and communicate through bounded, single-producer/single-consumer queues to keep the end-to-end delay below 2 s.

4.0.1 Thread 1 — Wi-Fi Reception & Pre-processing. Figure 7 depicts the Wi-Fi audio data receive thread. Each 1027-byte Wi-Fi packet comprises a 2-byte header, a 1024-byte VPU, microphone mixed sample and a 1-byte sequence index. Upon arrival, the thread

(1) splits the packet into VPU and microphone byte streams,
(2) applies a fixed-point amplify + denoise kernel, and
(3) pushes the cleaned audio into two lock-free byte queues (VPU_Queue, Mic_Queue).

The index field lets the host detect gaps and issue `RESEND` requests, guaranteeing loss-free delivery.

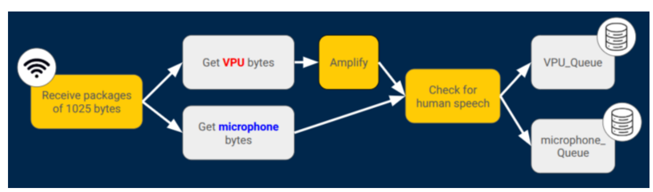

Figure 7: Thread 1: Wi-Fi reception and pre-processing.

4.0.2 Thread 2 — Voice Activity & Transcription. The transcription thread (Figure 8) drains bytes from VPU queues or Mic queue into a 10 seconds sliding window. For every new 2 seconds chunk of audio data the routine:

(1) **Window maintenance.** Once a Whisper inference finishes, the thread polls the VPU_Queue and Mic_Queue for a new 2 s audio chunk. If no data are available it enters a short sleep; otherwise, the chunk is appended (in order) to a 10 s sliding window. The window length is chosen because a spoken sentence typically spans 2–10 seconds: a full 10-s chunk would add prohibitive delay, whereas a 10-s *sliding* window

achieves a balanced trade-off between update latency and sentence completeness.

(2) **ASR inference.** The newest 10 s buffer is forwarded to *Whisper-medium*. After surveying hardware constraints, recognition accuracy and inference time as shown in **Table 2**, we selected the medium model as the best compromise for the target desktop GPU.

(3) **Result delivery.** The freshly generated sentence is pushed to the front-end Web application, where the transcript view is updated in real time.

Figure 8: Thread 2: Sliding-window transcription and UI update.

4.0.3 Thread 3 — User Interaction. A lightweight UI thread listens for `play`/`pause` events from the web dashboard and toggles the VAD gate accordingly, enabling hands-free start/stop control during meetings or interviews.

5 User Testing & Evaluation

We conducted three phased evaluations to assess SpeechSaw's performance under realistic conditions: (1) an internal pilot study to validate core functionality, (2) iterative refinement tests with controlled noise interference, and (3) an external test assessing real-world usability with 10 participants. These phases were designed to rigorously test the system's transcription accuracy and robustness to noise across realistic scenarios.

5.0.1 Internal Pilot Test. The pilot study aimed to validate core system functionality and to establish baseline comparisons between VPU-based input and conventional microphone capture. We conducted recordings using both the VPU and the microphone under two controlled acoustic environments:

- **Quiet Environment:** Soundproof booth with background noise capped at 30 dB.
- **Moderate Noise Interference:** YouTube video playback at 60 dB via external speakers.

Participants were instructed to read two standardized sentences (see Table 4) while recordings were simultaneously captured by both sensors. Transcriptions were generated using the Whisper-medium ASR model and evaluated using Word Error Rate (WER) as defined in Table 3. In the quiet condition, both sensors yielded comparable WERs, validating the integrity of the VPU's signal. However, in the moderate noise condition, the Whisper model's robustness limited our ability to differentiate sensor performance, suggesting the need for more challenging noise environments.

5.0.2 Refinement & External Test. Building on the pilot study insights, we designed a second round of testing with more aggressive and speech-relevant interference to simulate real-world scenarios. This phase introduced two additional noise conditions designed to challenge the Whisper model:

- **Noise Interference 1 – Alternating Speakers:** Pairs of users took turns reading alternating lines, simulating group discussion dynamics.
- **Noise Interference 2 – Overlapping Speech:** Multiple users spoke simultaneously, emulating crowded, multi-party environments.

The reading material was expanded to include 10 standardized test sentences (Table 5). For this phase, we conducted an external evaluation with 10 participants aged 18–25, grouped into pairs to simulate realistic multi-speaker environments. Each participant alternately wore the SpeechSaw prototype and performed the same set of speech tasks across three distinct noise conditions.

In both interference conditions, the VPU consistently outperformed the traditional microphone baseline (Table 7, 8, and 9). Notably, in the alternating speech scenario, the bone-conduction input of the VPU effectively attenuated ambient speech, resulting in a 30% relative reduction in WER. These findings highlight SpeechSaw's potential in noisy, privacy-sensitive environments such as meetings or shared workspaces.

Qualitative feedback revealed that while participants appreciated the compact, wrist-mounted design, several reported mild discomfort due to the tight fit required for optimal signal capture.

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A 3-D Model Assets

This appendix provides the two CAD models referenced in Section 3. Rendered views are shown in Fig. 9 and Fig. 10; the original STEP/STL files are included in the submission bundle[1].

[1]If the PDF viewer does not embed the attachments, they can be downloaded from our project repository.

Table 1: Key metrics across the three hardware iterations (datasheet values).

Prototype	Size (mm)	Active Power	Peak Data Rate	Notes
MCH Streamer	62×40	600 mW @ 5 V	480 Mbps (USB 2.0 HS)	miniDSP spec
ESP32-S3 DevKitC	57.1×27.9	340 mW (Wi-Fi RX, 103 mA @ 3.3 V)	150 Mbps (802.11 n)	Espressif DS
XIAO ESP32-S3	21×18	290 mW (Wi-Fi RX, 88 mA @ 3.3 V)	150 Mbps (802.11 n)	Seeed / Espressif DS

Table 2: OpenAI Whisper model variants and their resource requirements.

Model	Parameters (M)	VRAM (GB)	Relative speed*
tiny	39	~1	32×
base	74	~1	16×
small	244	~2	6×
medium	769	~5	2×
large	1 550	~10	1×

***Throughput relative to the *large* model (1×); higher is faster. Measured on a desktop-class GPU.**

Table 3: Common speech-recognition evaluation metrics.

Metric	Definition
Word Error Rate (WER)	$\frac{S+D+I}{H+S+D}$
Match Error Rate (MER)	$\frac{S+D+I}{H+S+D+I}$
Word Information Preserved (WIP)	$\frac{H}{\text{Length(Hypothesis)}} \cdot \frac{H}{\text{Length(Reference)}}$
Word Information Lost (WIL)	$1 - \text{WIP}$
Character Error Rate (CER)	WER ($\frac{S+D+I}{H+S+D}$) on a *character* basis

*H, S, D, **and** I **denote the numbers of** ***hits, substitutions, deletions,*** **and** ***insertions.***

Table 4: Reading material for the *designer first-step* test (internal pilot).

#	Sentence
1	"When the sunlight strikes raindrops in the air, they act as a prism and form a rainbow. The rainbow is a division of white light into many beautiful colors."
2	"These take the shape of a long round arch, with its path high above and its two ends apparently beyond the horizon."

Table 5: User-test reading resources. Resource 1 and Resource 2. For study test for 10 users

#	Resource 1	Resource 2
1	My name is Tom, what about you?	My watch fell in the water
2	I have a red pencil and eraser.	Prevailing wind from the east
3	This book is very interesting today.	Never too rich and never too thin
4	Today is Monday and it is sunny.	Breathing is difficult
5	I like to draw pictures every day.	I can see the rings on Saturn
6	She is my best friend today.	Physics and chemistry are hard
7	We are writing letters and numbers.	My bank account is overdrawn
8	The birds are singing in trees.	Elections bring out the best
9	I can ride my bicycle to school.	We are having spaghetti
10	He is reading a story tonight.	Time to go shopping

Table 6: Average Word-Error Rate (WER) across 10 users and 2 reading materials.

Condition	Sensor	WER
Single source	Mic	0.44
	VPU	0.47
Alternating sources	Mic	1.16
	VPU	0.57
Overlapping sources	Mic	0.61
	VPU	0.48

Notes.

(1) WER is computed as $(S + D + I)/N$, where S, D, and I are the numbers of substitutions, deletions, and insertions, and N is the reference length.

(2) Three acoustic scenarios were evaluated: (a) one user wearing a VPU speaks alone; (b) two speakers—one with a VPU, one without—take turns speaking; (c) the same two speakers talk simultaneously, producing overlapping speech.

Table 7: partial user test results in test condition 1, with reading the user study test resources 1

User	Input	Amplify dB	Predicted text	Avg dBFS	WER	MER	WIL	WIP	CER
1	classical	0	My watch fell in the water, prevailing wind from the east, never to reach and never too thin. Breathing is difficult. I can see the rings on Saturn. Physics and chemistry are hard. My bank account is overdrawn. Elections bring out the best. We are having spaghetti. Time to go shopping.	−57.24	0.41	0.41	0.65	0.35	0.08
	vpu	10	My watch fell in the water. The rolling wind from the east. Never too rich, and never too thin. Cleaning is difficult. I can see the rings on the salad. Physics and chemistry are hard. My bank account is really gone. Elections bring out the best. We are having spaghetti. Time to go shopping.	−30.15	0.47	0.44	0.67	0.33	0.17
	vpu	15	My watch fell in the water. The rolling wind from the east. Never too rich, never too thin. Breathing is physical. I can see the rings on the salad. Physics and chemistry are hard. My bank account is really long. Elections bring out the best. We are having spaghetti. Time to go shopping.	−34.60	0.47	0.44	0.67	0.33	0.17
	vpu	20	My watch fell in the water. The only way to peace. Never too rich, never too thin. Breathing is physical. I can see the rings on the salad. Physics and chemistry are hard. My bank account is illegal. Elections bring out the best. We are having spaghetti. I will go shopping.	−37.34	0.51	0.50	0.75	0.25	0.24
2	classical	0	My watch fell in the water. Preventing wind from the east. Never too rich and never too thin. Breathing is difficult. I can see the rain on the statue. Physics and chemistry are hard. My bank account is over drawing. Electronics brings out the best. We are having spaghetti. Time to go shopping.	−50.66	0.47	0.45	0.69	0.31	0.14
	vpu	10	My watch fell in the water. Preventing wind from the east. Never too rich and never too thin. Breathing is difficult. I can see the rain on the statue. Physics and chemistry are hard. My bank account is over drawing. Electronics brings out the best. We are having spaghetti. Time to go shopping.	−32.42	0.47	0.45	0.69	0.31	0.15
	vpu	15	My watch fell in the water. Preventing wind from the east. Never too rich and never too thin. Breathing is difficult. I can see the rain on the statue. Physics and chemistry are hard. My bank account is over drawing. Electronics brings out the best. We are having spaghetti. Time to go shopping.	−37.97	0.47	0.45	0.69	0.31	0.15
	vpu	20	My watch fell in the water. Preventing wind from the east. Never too rich and never too thin. Breathing is difficult. I can see the rain on the statue. Physics and chemistry are hard. My bank account is over drawing. Electronics brings out the best. We are having spaghetti. Time to go shopping.	−42.67	0.47	0.45	0.69	0.31	0.15

Notes. (1) WER is computed as $(S + D + I)/N$, where S, D, I are substitutions, deletions and insertions; N is reference length. (2) Acoustic scenarios: (a) one speaker with VPU; (b) two speakers (one with VPU, one without) speaking alternately; (c) the same two speakers talking simultaneously (overlap).

Table 8: Partial user-test results under test condition 2 (reading resource 1).

User	Input	Amplify dB	Predicted text	Avg dBFS	WER	MER	WIL	WIP	CER
1	classical	0	My watch fell in the water. Preventing wind from the east. Never too rich and never too thin. Breathing is difficult. I can see the rings on Saturn. Physics and chemistry are hard. My bank account is overdrawn. Election brings out the best. We are having spaghetti. Time to go shopping.	−52.69	1.10	0.63	0.76	0.24	1.02
	vpu	10	My watch fell in the water, never too rich and never too thin. I can see the rings on the side of the phone.	−19.03	0.48	0.48	0.68	0.32	0.37
	vpu	15	My watch fell in the water, never too rich and never too thin. I can see the wrong.	−24.48	0.55	0.55	0.68	0.32	0.48
	vpu	20	My watch fell in the water, never too rich and never too fair. I can see the wrong. Forgetting.	−29.19	0.55	0.55	0.69	0.31	0.46
2	classical	0	My watch fell in the water. Prevailing wind from the east. Never too rich and never too thin. Breathing is difficult. I can see the rain on the spectrum. Physics and chemistry are hard. My bank account is over drawing. Elections bring out the best. We are having spaghetti. Time to go shopping.	−52.58	1.21	0.66	0.79	0.21	1.11
	vpu	10	I wash out in the water. Never too thin. I can see the rain on Saturday. Bank account is over having spaghetti.	−39.76	0.66	0.66	0.84	0.16	0.36
	vpu	15	My watch showing never to reach and never to think. I can see the rain on Saturday. Bank account is over having spaghetti.	−44.29	0.62	0.62	0.82	0.18	0.31
	vpu	20	My watch showing never to reach and never to think. I can see the rain on Saturday. I'm drawing. I'm having spaghetti.	−47.51	0.69	0.69	0.87	0.13	0.40

Notes. (1) WER is computed as $(S+D+I)/N$, where S, D, I are substitutions, deletions and insertions; N is reference length. (2) Acoustic scenarios: (a) one speaker with VPU; (b) two speakers (one with VPU, one without) speaking alternately; (c) the same two speakers talking simultaneously (overlap).

Figure 9: 3D print model to contain the prototype board with esp32s3.

Table 9: Partial user-test results under test condition 3 (reading resource 1).

User	Input	Noise	Predicted text	Avg dBFS	WER	MER	WIL	WIP	CER
1	classical	0	My watch fell in the water, prevailing wind from the east. Never too rich and never too thin. Breathing is difficult. I can see the rings on siren. Physics and chemistry are hard. My bank account is overdrawn. Collections spring out the best. We are having spaghetti. Time to go shopping.	−51.82	0.39	0.39	0.63	0.37	0.14
	vpu	0	My watch fell in the water, prevailing wind from the east. Never too rich and never too thin. Breathing is difficult. I can see the rings on Saturn. Physics and chemistry are hard. My bank account is overrun. Elections bring out the best. We are having spaghetti. Time to go shopping.	−34.13	0.37	0.37	0.61	0.39	0.08
	vpu	1	My watch fell in the water, prevailing wind from the east. Never too rich and never too thin. Breathing is difficult. I can see the rings on Saturn. Physics and chemistry are hard. My bank account is overrun. Elections bring out the best. We are having spaghetti. Time to go shopping.	−38.69	0.37	0.37	0.61	0.39	0.08
	vpu	2	My watch fell in the water, prevailing wind from the east. Never too rich and never too thin. Breathing is difficult. I can see the rings on Saturn. Physics and chemistry are hard. My bank account is overrun. Elections bring out the best. We are having spaghetti. Time to go shopping.	−41.59	0.37	0.37	0.61	0.39	0.08
2	classical	0	My watch is showing the water. We are having spaghetti. Elections spring out the best. My bank account is overdrawn. Physics and chemistry are hard. I can see the rains are hard. My bank account is overdrawn. Elections spring out the best. We are having spaghetti. My watch falls in the water.	−49.4	0.82	0.79	0.95	0.05	1.55
	vpu	0	My watch fell in the water. Cream and new wind from the east. Never to reach and never to sing. Breathing is difficult. I can see the rain on the statue. Feast and chemistry are hard. My bank account is overdue. The luxury drinks are the best. We are having spaghetti. Time to go shopping.	−40.74	0.59	0.55	0.78	0.22	0.44
	vpu	1	My watch fell in the water. Cream and new wind from the east. Never to reach and never to think. Breathing is difficult. I can see the rain on the statue. Feast and chemistry are hard. My bank account is overdue. The lecturing brings out the best. We are having spaghetti. Time to go shopping.	−44.87	0.57	0.53	0.76	0.24	0.21
	vpu	2	My watch fell in the water. Creeping wind from the east. Never to reach and never to think. Breathing is difficult. I can see the rain on the statue. Peace and chemistry are hard. My bank account is overdue. The lecturing brings out the best. We are having spaghetti. Time to go shopping.	−47.64	0.53	0.51	0.75	0.25	0.19

Notes. (1) WER is computed as $(S + D + I)/N$, where S, D, I are substitutions, deletions and insertions; N is reference length. (2) Acoustic scenarios: (a) one speaker with VPU; (b) two speakers (one with VPU, one without) speaking alternately; (c) the same two speakers talking simultaneously (overlap).

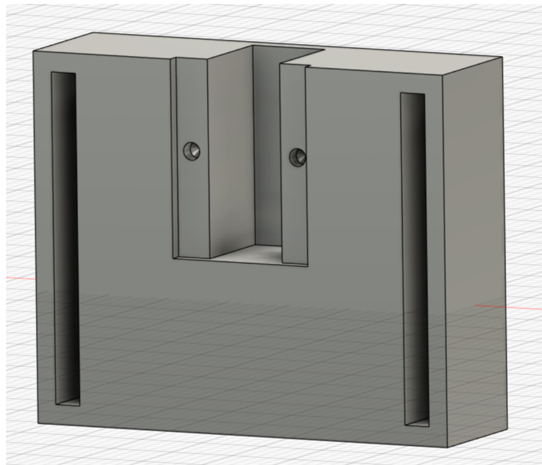

Figure 10: 3-D printed VPU wrist-housing

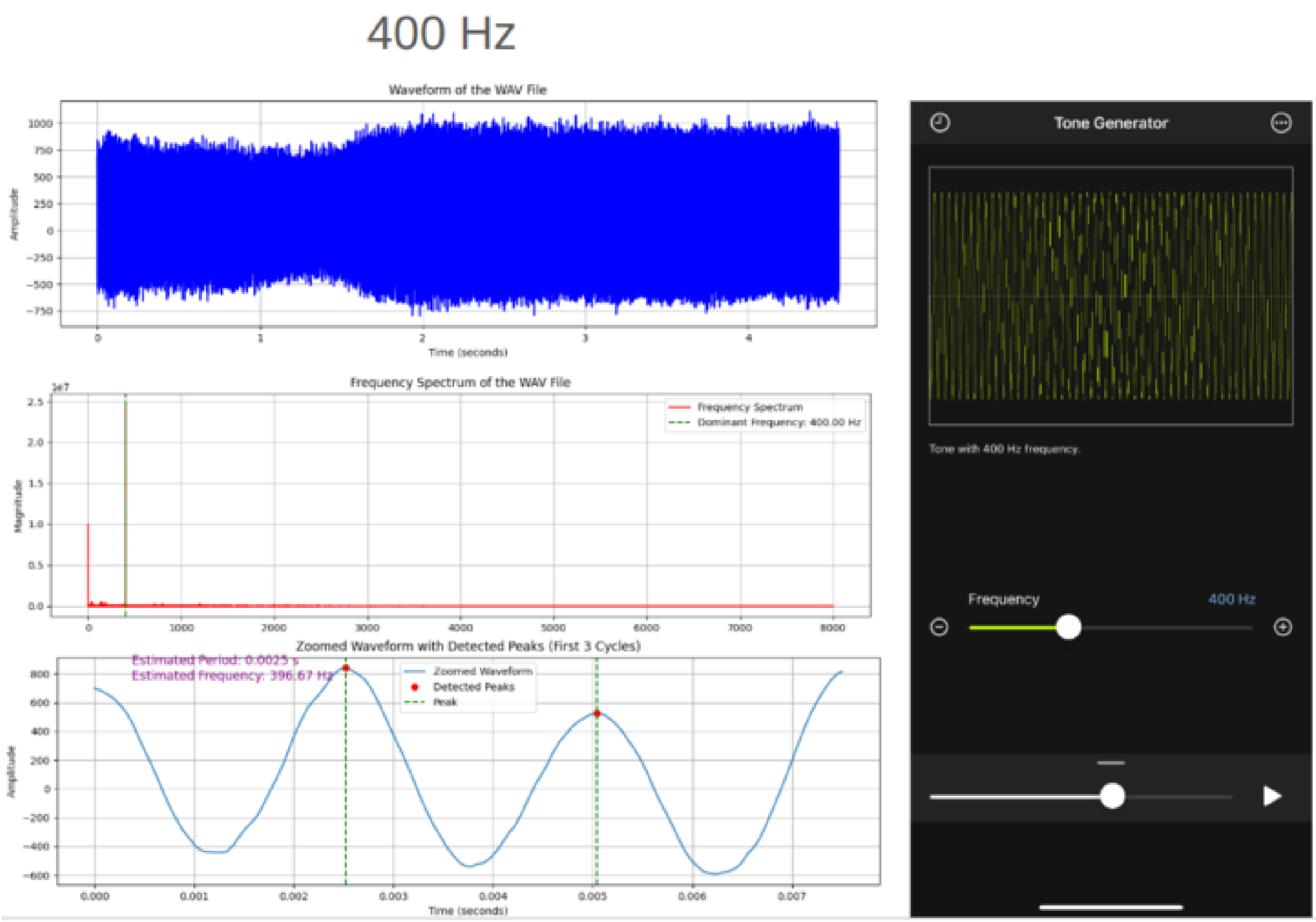

Figure 11: Caption text here.

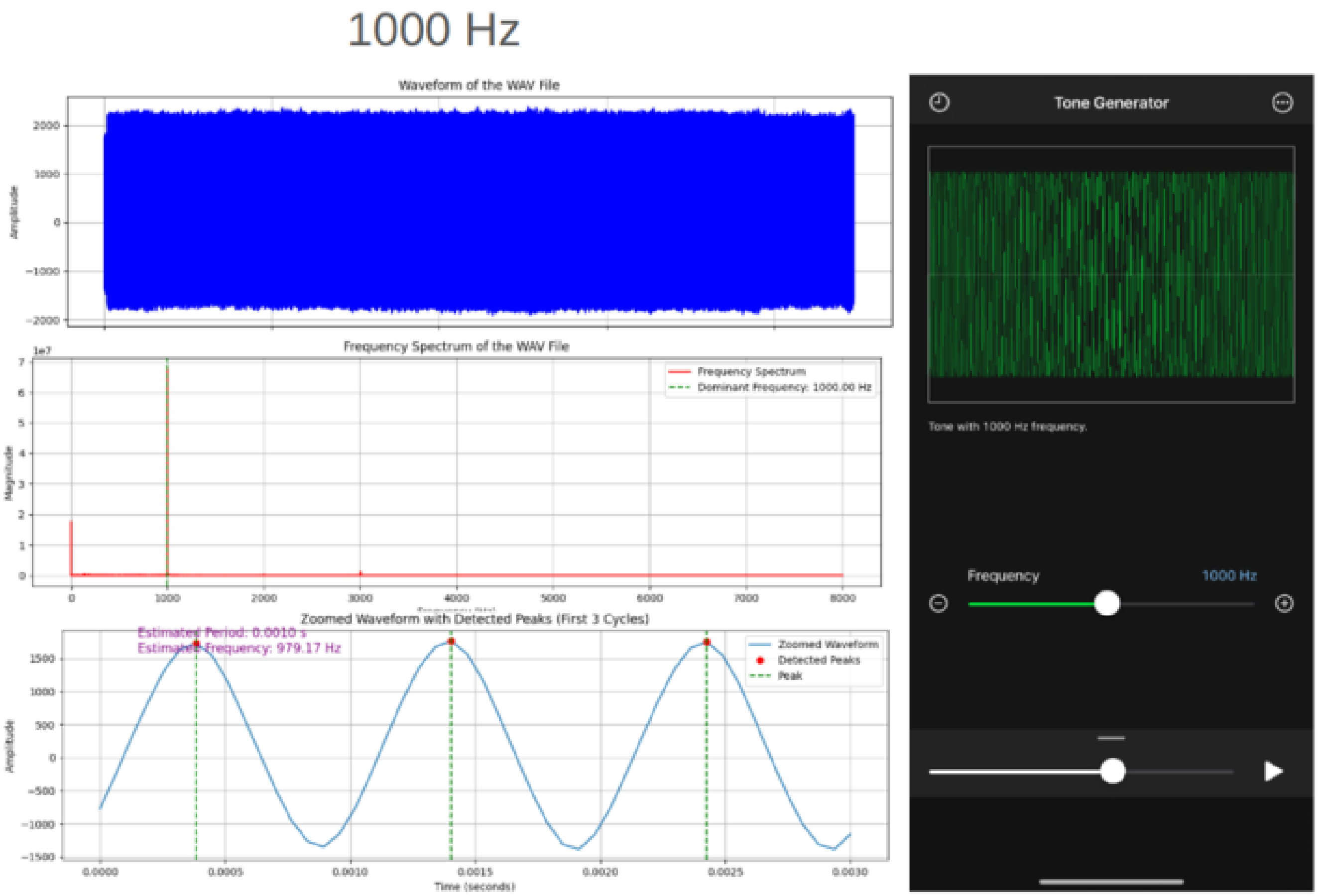

Figure 12: Caption text here.

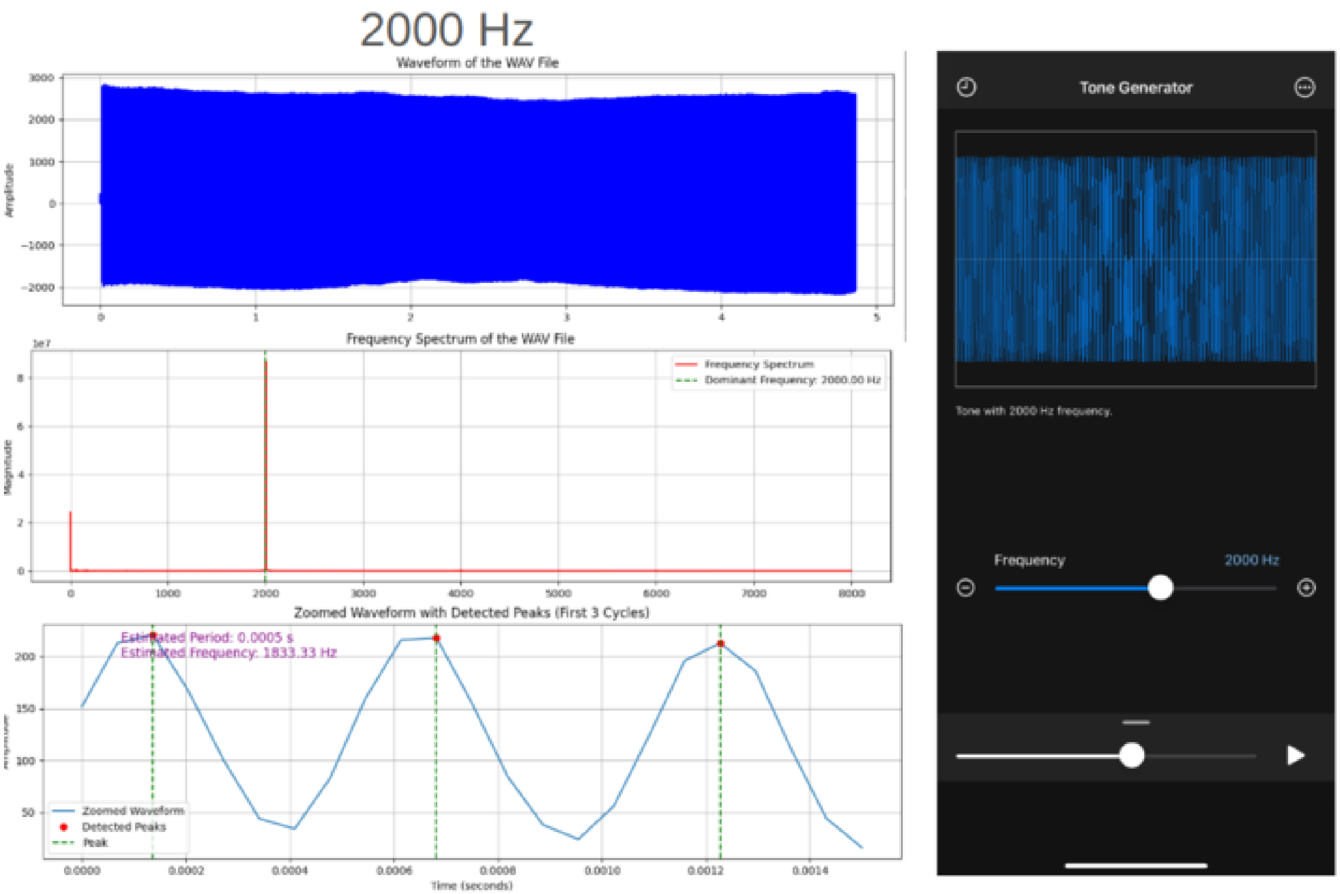

Figure 13: Caption text here.